Workflow Map: Current-State Supply Chain

A structured view of how materials move from supplier selection through receiving, laboratory use, and financial close-out. The map highlights ownership, handoffs, and the controls needed to keep materials traceable and costs accurate.

1. Workflow Map

Workflow stage	Key activities	Team(s) involved	Important handoff	Trigger / output
Stage 1: Procurement	Identify the need; define specifications; select and approve a supplier; create and issue the purchase order (PO).	Procurement; requesting department; approver	The approved PO and specifications are sent to the supplier and shared with the receiving team.	Trigger: approved material request. **Output:** approved PO and expected delivery.
Stage 2: Warehouse Receiving & Storage	Receive the shipment; verify it against the PO and delivery note; inspect quantity and condition; record the receipt; label and store materials correctly.	Warehouse Operations; Procurement; Quality / Laboratory, where inspection is required	Accepted materials and lot / batch details are recorded in the system and made available to the laboratory. Discrepancies are escalated.	Trigger: supplier delivery arrives. **Output:** recorded receipt, storage location, and acceptance or discrepancy record.
Stage 3: Laboratory Usage	Request and issue materials; record quantities consumed; link materials and lot numbers to the relevant experiment or work order; document results and remaining stock.	Laboratory Operations; Warehouse Operations	Actual material consumption and experiment / work-order references are recorded and passed to Finance for costing.	Trigger: approved laboratory request. **Output:** traceable material usage and experiment records.
Stage 4: Finance & Billing	Match the supplier invoice to the PO and receipt; review discrepancies; allocate material costs to the relevant work order or experiment; prepare billing or internal cost allocation.	Finance; Procurement; Warehouse; Laboratory Operations	Verified quantities, prices, and usage records support accurate cost allocation and any required customer billing.	Trigger: invoice received and/or usage records completed. **Output:** reconciled invoice, cost allocation, and billing record.
End State	Confirm that the material journey is fully documented, from purchase through receipt and use to financial reconciliation.	Procurement; Warehouse; Laboratory; Finance	All relevant records are linked and available for audit; exceptions are resolved or formally documented.	**Completion:** full material traceability, accurate costs, and an auditable record.

Core control: every handoff should preserve the material identifier, lot / batch number, quantity, status, and the relevant PO, receipt, experiment, or work-order reference.

2. Critical Dependencies

These dependencies connect the stages. If one is missing, the next team may be unable to proceed reliably.

Dependency 1: Procurement → Warehouse

Dependency: The warehouse needs an approved PO, supplier details, item specifications, and expected quantities before it can verify a delivery.

Potential impact if it fails: If the PO is missing or incorrect, receiving may be delayed, the wrong material may be accepted, or a discrepancy may go unresolved.

Dependency 2: Warehouse → Laboratory

Dependency: The laboratory depends on materials being received, inspected where required, labelled, and recorded with usable lot / batch information.

Potential impact if it fails: If receipt or traceability data is incomplete, laboratory work may be delayed and results may not be linked to the materials used.

Dependency 3: Laboratory → Finance

Dependency: Finance needs reliable consumption records linked to the relevant experiment or work order, as well as purchasing and receipt records.

Potential impact if it fails: If usage is not recorded accurately, costs may be allocated incorrectly, customer billing may be delayed, and the audit trail may be incomplete.

3. Potential Failure Points

Failure point	Likely consequence	Preventive / corrective control
PO details do not match the delivery.	Receiving delays, incorrect acceptance, or invoice mismatch.	Check each delivery against the PO and delivery note; record and escalate discrepancies before closing receipt.
Materials are stored without clear labels or lot / batch data.	Loss of traceability; material may be unavailable or unsuitable for an experiment.	Apply a consistent labelling standard and record the storage location and status in the inventory system.
Laboratory consumption is not recorded promptly.	Inventory records become inaccurate and experiment costs may be understated or overstated.	Record consumption at the point of use and require a link to the experiment or work order.
Invoice, receipt, and usage records cannot be reconciled.	Incorrect cost allocation, delayed billing, or audit exceptions.	Use a three-way match for PO, goods receipt, and invoice; investigate variances and retain supporting evidence.

4. Success Checklist

- Every material request has an approved purchase order and clear specifications.
- Every delivery is checked against the PO and recorded, including discrepancies.
- Materials are labelled, stored correctly, and traceable by lot / batch number.
- Laboratory usage is recorded against the correct experiment or work order.
- Finance can reconcile the PO, receipt, invoice, and usage records.
- Exceptions are assigned to an owner and resolved or formally documented.

- The full material journey can be reconstructed from records without relying on memory or informal messages.

Definition of success: materials are procured, received, stored, used, and financially reconciled with clear ownership at each handoff and a complete audit trail.